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Review

Environmental antimicrobial resistance and its drivers: a potential threat to public health

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ABSTRACT

Imprudent and overuse of clinically relevant antibiotics in agriculture, veterinary and medical sectors contribute to the global epidemic increase in antimicrobial resistance (AMR). There is a growing concern among researchers and stakeholders that the environment acts as an AMR reservoir and plays a key role in the dissemination of antimicrobial resistance genes (ARGs). Various drivers are contributing factors to the spread of antibiotic-resistant bacteria and their ARGs either directly through antimicrobial drug use in health care, agriculture/livestock and the environment or antibiotic residues released from various domestic settings. Resistant micro-organisms and their resistance genes enter the soil, air, water and sediments through various routes or hotspots such as hospital wastewater, agricultural waste or wastewater treatment plants. Global mitigation strategies primarily involve the identification of high-risk environments that are responsible for the evolution and spread of resistance. Subsequently, AMR transmission is affected by the standards of infection control, sanitation, access to clean water, access to assured quality antimicrobials and diagnostics, travel and migration. This review provides a brief description of AMR as a global concern and the possible contribution of different environmental drivers to the transmission of antibiotic-resistant bacteria or ARGs through various mechanisms. We also aim to highlight the key knowledge gaps that hinder environmental regulators and mitigation strategies in delivering environmental protection against AMR.

1. Introduction

Antimicrobial resistance (AMR) has emerged in humans, animals and the environment as a serious global concern in the 21st century [1,2]. Excessive use of antibiotics in medical, veterinary and agriculture sectors might be the root cause of worldwide AMR development. Injudicious sale of antibiotics over-the-counter, improper sanitation, and discharge of non-metabolised antibiotics or their residues through faeces/manure and industrial effluents into the environment have worsened the problem. There is a considerable relationship among these contributing factors and the environmental spread of AMR. Infections with multidrug-resistant microbial pathogens lead to approximately 700 000 annual deaths. If an effective action plan is not implemented, the annual death rate is expected to reach 10 million deaths per year by 2050, more than the deaths caused by cancer [3]. Recently, antibiotic consumption in cattle and poultry has risen at an unprecedented rate across continents, which is expected to rise further by 67% by the year 2030 in rapidly polluting and developing countries of the world [4]. The growing global issue of AMR has also significantly contributed to the world's economic healthcare load. Measurement of the global cost of antibiotic resistance is difficult, but indisputably the AMR-related financial burden is considerably high [5].

The misuse or abuse of antibiotics generates unnatural selective pressure in clinical and natural environments that is detrimental to human and animal health [6]. The evolution of resistance in bacteria is a natural process. The development of resistance occurs genetically either by mutation or acquisition of new resistance genes through genetic exchange mechanisms [7,8]. Antibiotics are well known to stimulate the resistance mechanism of pathogens through horizontal gene transfer. Currently, microbial resistance to third-generation cephalosporins, carbapenems and polymyxins is increasing rapidly [9,10]. Resistance of bacterial pathogens to third-generation cephalosporins is related to the acquisition of

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extended-spectrum β-lactamase (ESBL) genes. According to one estimate, ~14% of the global human population is colonised by ESBL-producing enteric bacteria and it is growing at a rate of 5.4% annually [11]. In 2017, the World Health Organization (WHO) has reported resistance to third-generation cephalosporins, carbapenems, clindamycin and the last-resort antibiotic vancomycin. Increasing microbial resistance to last-resort antibiotics has urged the WHO to sponsor further research for the development of novel antimicrobial lead molecules against priority pathogens [12]. There are rising concerns about the last group of β-lactam (carbapenem) and polymyxin (colistin) antibiotics that possess broad-spectrum activity against Gram-negative bacteria. The first incidence of the New Delhi metallo-β-lactamase type 1 (NDM-1) gene in a clinical patient was reported in 2008. Subsequently, this gene has been identified worldwide and nowadays is often detected in various environmental samples [13]. Likewise, resistance to colistin attributed to the plasmid-mediated mobile colistin resistance *mcr-1* gene was first reported in 2016 but is now found worldwide [14,15]. Non-antibiotic antimicrobial compounds (metals and biocides) prevalent in environmental samples are also responsible for the selection of resistance. Zinc-, copper-, mercury- and nickel-mediated selection of antimicrobial resistance genes (ARGs) has been well reported. The application of biocides, herbicides, pharmaceuticals and pesticides has also led to the selection of resistant micro-organisms, which pose serious challenges to the natural environment [16,17].

Leading world health agencies and surveillance systems track AMR as a paramount global issue owing to expanding animal and human populations, international trade, increased globalisation, increased demand for animal food, and too-easy access to antimicrobial compounds in developing and developed countries [18]. Prolonged exposure to subtherapeutic doses of antimicrobials used in animal production provides ideal conditions for bacteria to fix their resistance-related genes. Such resistance genes can be transmitted to the human gut microbiota via contaminated foodstuffs or the environment [19,20]. Studies of transmission dynamics provide an in-depth understanding of the transmission of resistant pathogens among humans [21]. Prolonged hospital stay and unhygienic medical conditions lead to the transmission of community-acquired methicillin-resistant *Staphylococcus aureus* (MRSA) [22]. Outrageous use of antimicrobial compounds as growth promoters in poultry or livestock is another important cause of resistance transmission to humans through animal products. The most common resistant pathogens in this category are *Campylobacter* spp., *Salmonella* spp. and food-borne *Escherichia coli* [23–25]. Surveillance systems should adopt a global 'One Health' approach, which involves the interdisciplinary and collaborative efforts in achieving the optimal health of humans, animals and the surrounding environment, to counter this threat [26]. Mendelson and Matsoso have categorised the prevention, diagnosis and treatment of infectious diseases into three broad categories [27]. The first step is to prevent the easy accessibility of antibiotics locally, socially and globally. Second, conservation activities should be followed to minimise the use of antimicrobials through appropriate prescriptions, infection control and surveillance. The third step is to foster innovative biotechnological advancement in infection control technologies, diagnosis and the production of next-generation antimicrobials or vaccines [27]. Recent reports suggest that AMR action plans lack a necessary element of involving all environmental regulators and risk assessments to enhance the efficacy of existing antimicrobials for future applications [28–30]. This review briefly highlights the epidemiology of AMR in human and veterinary practices, environmental drivers and their significance in AMR development, as well as key knowledge gaps for better understanding of the environmental spread of resistance. These factors might offer new insights into combating AMR-related issues.

2. Antimicrobial resistance: a global crisis

AMR is a worldwide problem that primarily derives from antibiotics and antibiotic resistance genes, but inadequate local sanitation, pollution and other factors also play an important role [5]. Major causes of antibiotic resistance include imprecise usage and inadequate regulations, lack of awareness leading to excessive or inept antibiotic use, poor sanitation/hygiene, the release of non-metabolised antibiotics or their residues into the environment through manure/faeces, and the use of antibiotics as growth promoters in poultry and livestock rather than as infection controlling agents [7]. These factors contribute to genetic selection pressure for the emergence of multidrug-resistant bacterial infections in the community. *Salmonella* spp. and *Campylobacter* spp. are the major pathogens in this category. Additionally, indistinguishable resistance mechanisms have been found in bacteria isolated from humans and animals. The study of resistant microbes should cover a range of factors, including the evolution of resistance at the molecular level within a given organism, transmission mechanisms and pathways between organisms, and dissemination between humans and animal hosts and across the wider environment including soil and water. Approximately 80% of marketed antibiotics in the USA are used either as growth supplements or to control animal infections. The global map of antibiotic use estimated the use of 63 151 tonnes of antibiotics in livestock in 2010 [4]. This situation would pose a worldwide economic burden of approximately US$120 trillion (US$3 trillion per annum), which is roughly equal to the total current annual healthcare budget of the USA. Recent trends of antibiotic usage will lead to the death of approximately 444 million people before 2100, and birth rates will also rapidly decline by 2050 [31].

3. History and epidemiology of antimicrobial resistance in human and veterinary practices

Antimicrobial compounds are used in a variety of environmental settings such as hospitals, intensive care units, outpatient clinics, animal farms, feedlots and veterinary clinics, which are increasing the burden of resistance emergence [32]. The discovery and development of antibiotics for combating infectious diseases is the biggest advancement in the history of chemotherapy [33]. However, indiscriminate and extensive usage of antibiotics during the last 70 years has resulted in the selection and emergence of resistant pathogens against almost every antibiotic that has been launched to date [7]. In clinical practice, penicillin resistance was identified in *Streptococcus pneumoniae* and *S. aureus* in 1940, which demonstrated the presence of previous penicillin resistance mechanisms in the natural environment before its clinical use [34]. Likewise, after the introduction of methicillin, a semi-synthetic penicillin, for treating infections by penicillin-resistant *S. aureus*, a methicillin resistance mechanism was also recorded in strains of MRSA. *Staphylococcus aureus* is closely associated with humans as it is a nasal commensal in ~30% of the population and causes a skin infection known as boils. Recently, MRSA has become a major community-acquired pathogen with enhanced virulence and transmission characteristics to become a serious nosocomial pathogen [35–37]. Antibiotic-resistant Gram-negative bacteria, mainly members of the Enterobacteriaceae family, have been increasingly reported from hospital wastewater [38–40]. These reports describe that the use of almost every antibiotic eventually results in the development of resistant strains, proving the extreme flexibility, hostility and wide-range adaptability of the bacterial genome. The use in food animals of antibiotics that are structurally similar to agents used for humans exacerbates antibiotic-resistant bacteria of public-health concern. Enteric antibiotic-resistant bacteria are ex-

Table 1
Major groups of problematic multidrug-resistant bacteria and their resistance genes from various environmental samples

Pathogen	Source of isolation	Antimicrobial resistance genes	Responsible for	Reference(s)
Escherichia coli	Household water, drinking water, poultry swabs, milk, vegetables, raw meat and chickens	*bla*$_{CTX-M-9}$, *qnrS*, *bla*$_{CTX}$, *bla*$_{TEM}$, *bla*$_{CTX-M}$ + *bla*$_{SHV}$, *bla*$_{SHV}$, *bla*$_{CTX-M}$ + *bla*$_{TEM}$, *bla*$_{TEM}$ + *bla*$_{SHV}$, *bla*$_{CTX-M}$ + *bla*$_{SHV}$ + *bla*$_{TEM}$, *bla*$_{CMY-2}$	Urinary tract infections, diarrhoea, neonatal meningitis, septicaemia, abdominal pain, pneumonia, nosocomial bacteraemia	[46–49]
Staphylococcus aureus	Retail meat and broiler chickens	*mecA, tetK, blaZ*	Bovine mastitis, osteomyelitis, endocarditis, gastroenteritis, toxin syndromes, abscess formation	[50,51]
Klebsiella pneumoniae, Klebsiella oxytoca	Milk	*aadA22, arr-3, aac(3)-Id, cmlA*, *bla*$_{TEM}$, *ereA2*, *bla*$_{SHV}$, *bla*$_{OXA}$, *bla*$_{CTX-M}$	Urinary tract infections, bovine mastitis, septic shock, pulmonary pneumonia, wound infections, nosocomial infections, hepatic and pancreatic disease, septicaemia, bacteraemia	[52,53]
Enterococcus faecium	Chicken faeces and water	*tetM, ermB, tetL*	Vancomycin-resistant enterococci (VRE) infections, neonatal meningitis nosocomial infections	[54]
Salmonella spp.	Retail pork, pigs, broiler chickens	*pse-1, tetA, tetB, ant(3′′)-Ia*	Salmonellosis, bacteraemia	[55,56]
Campylobacter spp.	Chickens	NA	Post-infectious irritable bowel syndrome, campylobacteriosis	[57,58]
Pseudomonas spp.	Untreated and treated water	*bla*$_{AmpC}$, *bla*$_{OXA}$, *bla*$_{IMP}$, *bla*$_{TEM}$, *tetC, tetA, sul1*	Blood-borne infections, central nervous system infections, urinary tract infection, wound infections, endocarditis, nosocomial infections	[59,60]
Vibrio spp.	Urban water, shell fish	*dfrA15*, *bla*$_{P1}$	Cholera, diarrhoea, profuse watery stools	[61,62]

NA, not available.

creted in faeces and can disseminate in the environment and possibly enter other livestock and farmworkers. Resistant bacteria and mobile genetic elements may enter human food from different animal sources, as all processing stages contain abundant resistant bacteria and resistance genes [41,42]. Members of the Enterobacteriaceae are frequently distributed in livestock and on retail meat, including opportunistic pathogens such as *Escherichia* and *Klebsiella* spp. These pathogens commonly cause clinical urinary tract infections and bloodstream infections in patients [43].

Some ARGs found in foodborne pathogens have also been identified in humans, thus confirming the risk of consuming animal products colonised by resistant bacteria [44,45]. Major groups of multidrug-resistant pathogens and their resistance genes identified from various environmental samples are presented in Table 1. A study of Vietnamese chicken farms revealed that 84% of antibiotics were used for prophylaxis, 12% in disease management and 3.8% as combined use for prophylaxis and disease management [63]. The increase in food-animal consumption with the rising economies of low- and middle-income countries will probably increase antibiotic consumption until it exceeds the present-day limit of high-income countries. The Organization for Economic Cooperation and Development (OECD) has suggested that antimicrobial usage in food animals will rise from 63 151 tons in 2010 to 105 596 tons by 2030 [4]. The average annual global consumption of antibiotics in producing 1 kg of different meats is 172 mg/kg for pork meat, 148 mg/kg for chicken and 45 mg/kg for beef [4]. This consumption is speculated to double in Brazil, Russia, India, China and South Africa (BRICS) because of the increasing requirement of antibiotics to maintain animal health and productivity [4]. The contribution of five major countries in food-animal production is depicted in the form of a pie chart in Fig 1.

4. Common drivers of antimicrobial resistance

A considerable amount of research data have indicated that there are well-characterised classes of resistance-driving compounds or drivers that confer AMR, including: (i) antimicrobials such as antibacterial, antifungal, antiviral and antiparasitic drugs; (ii) heavy metals; and (iii) biocides, mainly surfactants and disinfectants. Resistance to natural compounds (plant-derived) and xenobiotics (hexane, toluene and octanol) has also been reported to select for resistance genes [64,65]. Co-selection of resistance genes has been reported for hazardous chemicals such as solvents [66], biocides [67,68], heavy metals [69] and antibiotics. There are two different means of co-selection such as: (i) co-resistance, where selection of one gene supports the selection of another gene that usually does not offer a selective advantage to the compound of interest [70,71]; and (ii) cross-resistance, where one resistance gene protects against a range of toxic chemicals [72,73]. The major transmission pathways of these drivers in the environment include (i) industrial and municipal wastewater, (ii) sewage sludge and spreading of animal manure and (iii) aquaculture systems. Some of these drivers and their role in AMR are described briefly below.

4.1. Resistance driver: antibiotics and antibiotic resistance genes

4.1.1. Consumption of antibiotics

Since the discovery of penicillin, antibiotics have played a significant role as therapeutic agents for the treatment of various human infectious diseases, and some of these antibiotics are also commonly used in the aquaculture and livestock industries [74]. Multidrug-resistant (opportunistic) pathogens such as *Acinetobacter baumannii* and *Pseudomonas aeruginosa* may grow in niches where other bacteria cannot survive and may also disturb the host microbiota. Antibiotic misuse and overuse result in the selection of resistant strains. An estimated 50% of antibiotic prescriptions have been suggested to be inappropriate. In many countries outside of Europe and North America, non-prescription use of antibiotics is still common, accounting for 19% of total consumption and even up to 100% in some cases [75]. Approximately 20–30% of European patients receive antibiotics during their hospitalisation, and hospital effluents are hotspots for horizontal gene transfer, which can facilitate the interspecies and intraspecies transfer of antibiotic resistance determinants and virulence factors [76]. The existence of mobile genetic elements (transposons and integron-associated gene cassettes) facilitates horizontal gene transfer, which can efficiently contribute to the acquisition, maintenance and spreading of ARGs within bacterial communities [77]. Evaluation of ARGs in clinical and envi-

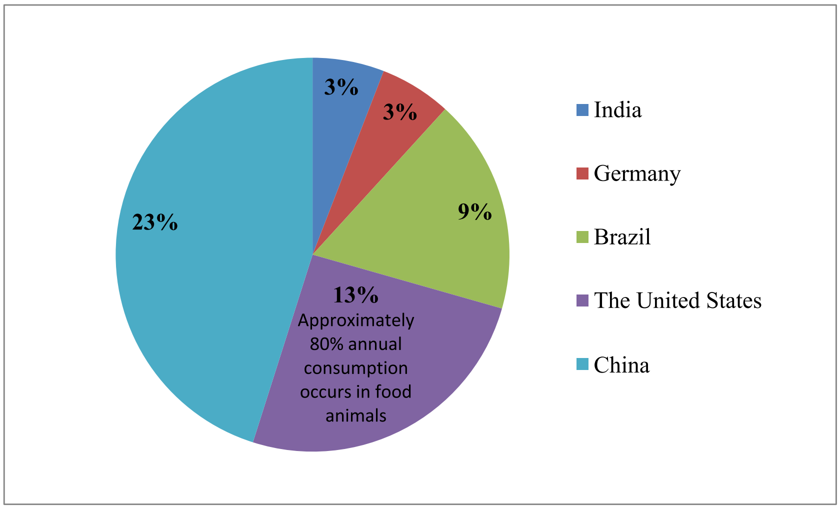

Fig. 1. Five major countries involved in the greatest share of antimicrobial consumption during food-animal production [148].

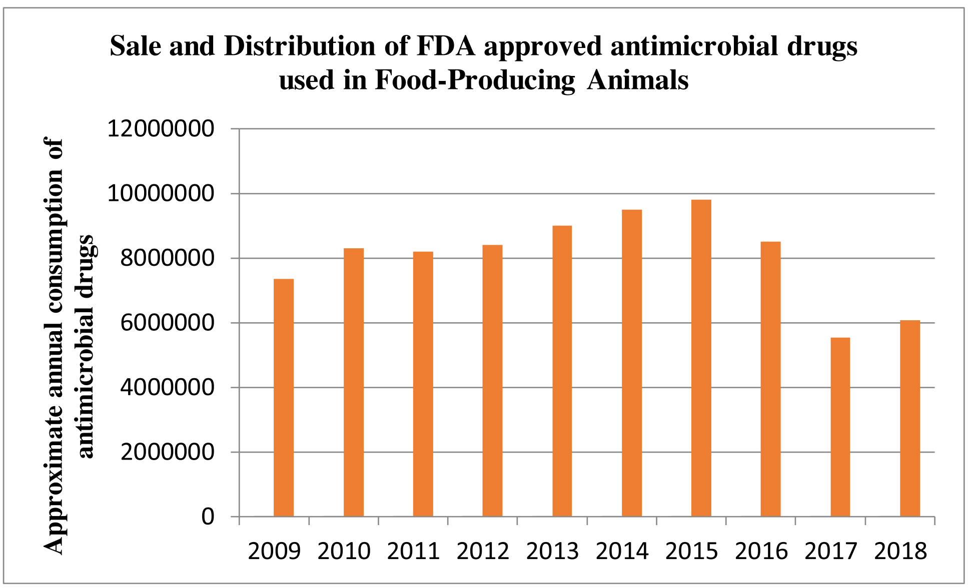

Fig. 2. Sales and distribution of US Food and Drug Administration (FDA)-approved antimicrobial drugs (modified from FDA, 2018 summary report).

ronmental settings should be the first step in tackling the rapidly growing resistance to antibiotics.

Veterinary medicines were intended to minimise animal infections but are currently used as prophylactic agents, feed additives and growth promoters [78]. India is the fourth largest producer of antimicrobials for animals after China, the USA and Brazil. Antibiotic residues have been identified in food-animal products such as chicken, meat and milk [79]. According to the US Food and Drug Administration (FDA) annual summary report on the sale and distribution of antimicrobials in food-producing animals, consumption of medically important antimicrobial drugs has increased by 9% from 2017 to 2018. The year 2018 was the second lowest year with a 21% decrease in antimicrobial consumption since the first year of the report (2009) and a 38% decrease compared with the peak year of sales and distribution (2015) (Fig. 2). A detailed account of antibiotics used in animals can be found in the Consensus Statement by Morley et al. [80]. Similar to humans, 30–90% of antibiotics consumed in animals are released into manure and urine. Due to the high rate of antibiotics on animal farms, a high concentration of antibiotic residues in farm waste contaminates the environment with antibiotic-resistant bacteria [81]. This situation further facilitates the selection and transfer of antibiotic resistance genes within the microbial community of an environment. These resistance genes are transferred to humans and other environments through the food supply and waste materials [82,83]. *Escherichia coli* O157:H7 and *Salmonella* spp. have been detected in manure, compost, soil and water samples. *Salmonella* was identified in 6 of 380 manure samples from pig farms [84] and in 6 of 96 water samples from irrigation and non-irrigation sources of leafy green farms [85]. Xu et al. isolated *Salmonella* strains from breeder chickens in Henan (China) and 69.64% of strains were found to be resistant to three or more antimicrobials of different classes [86]. The rigorous use of antimicrobials in veterinary and medical fields has resulted in high AMR through the exertion of selection pressure against antimicrobial agents. Some of the commonly used antibiotics in animal husbandry for prophylactic purposes along with their negative impacts on animal and human health are listed in Table 2.

4.1.2. Antibiotic transmission pathways into the environment

AMR transmission can be distributed among humans, animals and the environment via different routes [21]. The ecosystem

Table 2
Consequences of commonly used antibiotics on human and animal health

Antibiotic	Antibiotic residues detected in	Impact on animal/human health	Reference(s)
Amoxicillin	Eggs, milk	Carcinogenic and mutagenic effects	[87]
Quinolones	Chicken and beef	Hypersensitivity reactions, phototoxic effects on skin and tendon rupture	[88,89]
Sulfonamides	Raw milk	Carcinogenicity and allergic reactions	[90]
Oxytetracycline	Raw milk, kidney, liver and muscles of beef and chicken, and muscles of calves	Carcinogenicity and cytotoxicity in broiler chicken bone marrow	[91]
Tetracycline	Liver and muscles of chicken, and liver, muscles and kidney of beef	Primary and permanent tooth discolouration in children and infants, nephrotoxicity, carcinogenicity, hepatotoxicity and alteration of the normal intestinal microflora, hyperpigmentation of skin exposed to the sun, acidosis in proximal and distal renal tubules	[92,93]
Sulfadimidine, sulfamethoxazole	Raw milk	Allergic reactions and carcinogenicity	[94]

serves as a bridge for various compartments of animals to soil to water to sand, and to sewage. The environment acts as a reservoir to mix mobile genetic elements that interact and spread to other parts or to human and animal hosts [95]. Discharge of antibiotics occurs through several routes such as municipal and hospital waste, animal husbandry, the manufacturing industry, runoff from agricultural fields containing livestock manure and landfill leachates of antibiotic discharge [30,96]. The half-life of antibiotics ranges from hours to hundreds of days but antibiotic residues are considered as persistent contaminants of the environment [97].

Increased use of antibiotics generates robust selective pressure on natural microbial ecosystems and humans [15]. Similar to other pharmaceutical products, unused or expired antibiotics are thrown in the garbage that cannot degrade and enter the groundwater or aquatic system during the wastewater treatment process. A very limited concentration of these disposed of antibiotics can promote selective pressure either by horizontal gene transfer or modifying target sites [98]. Primarily, such resistance genes originate from the gastrointestinal tract of humans [19]. Similarly, 70% of Gram-positive and 65% of Gram-negative bacterial isolates from a natural cave microbiome contained multiple active antibiotic resistance genes, suggesting that multiple resistance is an environmental standard instead of an exceptional response to elevated anthropogenic antibiotic stress [99]. Resistance genes have also been found in 30 000- to 5000-year-old samples of permafrost but comparatively there is a clear spike in the abundance of resistance genes in recent environmental samples [80]. Higher levels both of antibiotic-resistant bacteria and ARGs have been detected in various wastewater samples including municipal wastewater, sewage, and influents and effluents of wastewater treatment plants [100–102]. Likewise, high levels of AMR have been identified in agricultural and industrial wastewater, including samples from pharmaceutical sources [103]. Increased rates of ARGs have been identified in pharmaceutical wastewater associated with the rate of clinically important antibiotic drugs that are used during different treatment stages and are subsequently released into the environment. Hospitals and other healthcare facilities are essential sources of antimicrobial waste generation either indirectly by patient secretions or directly as discarded medicines. Diwan et al. recorded a troubling degree of fluoroquinolone, sulfonamide and tinidazole residues in hospital effluents [104]. Recent studies have demonstrated the presence of novel ARGs, enriched in activated sludge and effluent wastewater, that confer resistance to multiple antibiotics including aminoglycosides, fluoroquinolones and β-lactams [105]. Activated sludge has been found to possess ESBL-producing *E. coli*, MRSA and vancomycin-resistant enterococci containing *vanA* and *mecA* genes conferring resistance to vancomycin and β-lactams, respectively [106]. India and China are the largest contributors to the global rise of AMR, therefore strict surveillance is urgently required. A generalised scheme of different AMR drivers and their transmission to humans is represented in Fig. 3.

Unlike clinical micro-organisms, antibiotic-producing environmental micro-organisms reveal a higher degree of intrinsic resistance that appears to be completely distinct from selective pressure and is not acquired by horizontal gene transfer. The increasing risk of transferring intrinsic resistance prevalent in environmental micro-organisms to human-adapted pathogens is already evident and has become a major concern in healthcare settings [2,103]. Antibiotic resistance due to the irrational use of antibiotics, the increased frequency of antibiotic-resistant bacteria, and mechanisms of antibiotic resistance in the environment are now referred to as 'antibiotic resistance pollution' [107].

4.2. Resistance driver: heavy metals

Metal and antibiotic resistance mechanisms share some structural and functional similarities [108,109]. Similar to antibiotics, metals including arsenic, zinc, cobalt, manganese and silver also have resistance mechanisms such as: (i) reduced membrane permeability similar to β-lactams, ciprofloxacin, tetracycline and chloramphenicol; (ii) metal alteration, similar to chloramphenicol and β-lactams; (iii) efflux of metals such as zinc, nickel, copper, cobalt and cadmium, similar to chloramphenicol, tetracycline and β-lactams; (iv) target site modification for zinc, copper and mercury; and (v) metal sequestration for copper, cadmium and zinc, similar to coumermycin A [110].

Pal et al. reported that among 20 metal resistance genes, Cu resistance genes are the most commonly distributed following the decreasing abundance of non-essential metals in the BacMet database [111]. Considerable information is available regarding the toxicity and mobility of metal species in sediment, aquatic and soil ecosystems [36,112–114]. The desired goal to limit resistance selection within soil habitats requires revisiting the environmental thresholds of sludge and manure land spreading since the combined exposure of antibiotics, metals and biocides exceeds the MSC (minimum selection concentration) for some shared resistance mechanisms. Bacteria harbouring metal resistance genes have been reported to carry ARGs more frequently compared with those without metal resistance genes, which are often located on plasmids [115].

4.3. Resistance driver: biocides

Biocidal products include an array of chemical compounds that exert microbiostatic or microbiocidal effects against a broad range of micro-organisms. Disinfectants are commonly used in cosmetics, hospitals, household cleaning products, wipes, and industrial processes including fouling management and souring of pipes [116]. In Europe, the marketing, usage and disposal of biocides

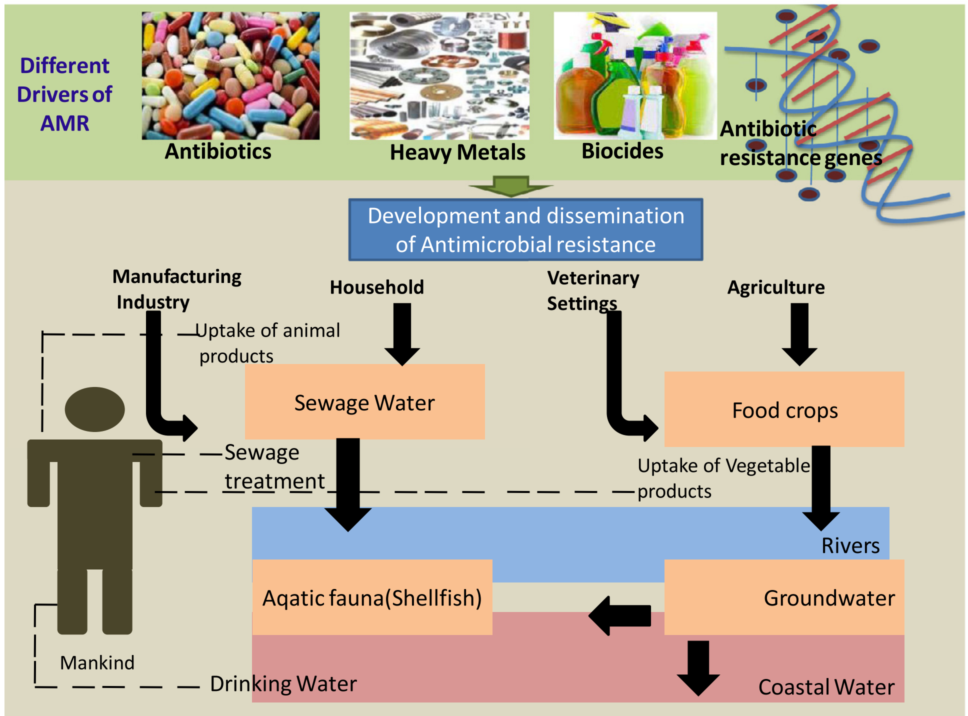

Fig. 3. Generalised scheme of different drivers of antimicrobial resistance (AMR) and their possible route of transmission to humans.

are controlled under the European Biocidal Product Regulation (EU 528/2012) [117]. This regulation also controls the use of biocides in veterinary settings. The most commonly used biocides include formaldehyde, chlorhexidine and quaternary ammonium compounds (QACs) [e.g. alkyldimethylbenzylammonium chloride (ADBAC), isothiazolium-benzalkonium chloride, ethanol, cetrimonium chloride/bromide (cetrimide), aminoalkyl glycines, triclosan, cetylpyridinium chloride and didecyldimethylammonium chloride (DDAC)] [118]. The minimum inhibitory concentration (MIC) of biocides has been determined against multidrug-resistant pathogens, ranging from 0.4–1000 µg/mL. Triclosan in combination with other biocides such as QACs and chlorhexidine has been found to select for antibiotic resistance in microbial pathogens [119]. Sublethal concentrations of biocides also facilitate the selection of mutations that confer antibiotic resistance, similar to the selection of ARGs at sublethal concentrations of antibiotics [120,121]. The co-existence of ARGs on the extrachromosomal genome improves their resilience and transmission throughout pathogens or microbial communities [70,122]. For example, co-resistance to the biocide benzalkonium chloride and antibiotics confers an eight-fold higher tolerance to *S. aureus* against antibiotics compared with the sensitive wild-type strain [122]. These genes are frequently co-located on plasmids. Recently, the antimicrobial compound triclosan and the antidepressant agent fluoxetine have been reported to mediate the selection of resistance. The profusion of these compounds has also been reported in many wastewater environments, which is raising concerns [123,124].

5. Impact of antimicrobial resistance on public health

The indiscriminate release of antibiotics into the environment is creating an economic burden on the global healthcare system. The European Centre for Disease Prevention and Control (ECDC) reported that almost two million people in the EU/EEA get infected with antibiotic-resistant bacteria every year leading to approximately 25 000 annual deaths [125]. Chronic infections, delayed treatment and transmission of resistance to other species are some of the AMR attributes. AMR and multidrug-resistant healthcare-associated infections have led to the rise of various clinical problems. The US Centers for Disease Control and Prevention (CDC) estimated that approximately 223 900 cases of *Clostridioides difficile*, 197 400 cases of ESBL-producing Enterobacteriaceae and 32 600 cases of multidrug-resistant *P. aeruginosa* occur in hospitalised patients in the USA every year [126]. Extensive consumption of antibiotics and risk assessment of antibiotic residues on human health should receive increasing attention in the current scenario. The hazard assessment of drug-resistant bacteria should include four main domains, including identification of hazards, exposure assessment, characterisation of risk and the dose–response relationship [127,128]. Antibiotic residues in the environment are considered to alter the human microbiome resulting in the emergence and selection of resistant bacteria inhabiting the human gut, and known as human antibiotic resistance [129]. Sometimes, environmental antibiotic resistance also results from the selective pressure on the environmental microbiome that acts as a pool of ARGs and antibiotic-resistant bacteria [130].

Gram-negative pathogens causing nosocomial (hospital-associated) infections include *P. aeruginosa, A. baumannii, C. difficile, Burkholderia cepacia* as well as pathogens such as *Campylobacter jejuni, E. coli, Haemophilus influenzae, Klebsiella pneumoniae* and *Salmonella* spp. These pathogens cause several diseases in humans and animals [131].

The CDC estimated that each year 23 000 Americans die from antibiotic-resistant infections [18]. Many common infections are resistant to first-line antibiotics and the types of bacteria related to these infections are also found in livestock. Investigation of the association between the usage of antibiotics on farms and the human health risk is a complex process. The evolutionary dynamics of AMR do not follow a simple model for the emergence of resistant pathogens for epidemiological reasons [132]. The individual contribution of different antimicrobial drugs on the overall burden of AMR is still not clear.

6. Key fundamental gaps in antimicrobial resistance knowledge

Similar to climate change, the AMR issue is also considered interdisciplinary and of global concern [133]. Many years have been spent in generating the necessary knowledge and understanding of AMR mechanisms, selection and transmission. However, we still have little awareness about the fundamentals of the ecological impact of different AMR drivers in the environment. AMR has become a global threat and filling the knowledge gaps is urgently required on a priority basis to understand the potential role of selective agents in the evolutionary processes in the environment leading to resistance. Some knowledge gaps require risk assessments in humans and animals. Assessment of knowledge gaps and modelling can support the identification of hotspots of antibiotic resistance emergence and dissemination, which would be helpful in finding the mitigation targets. To achieve detailed knowledge of the key gaps in environmental AMR, readers can draw attention on various excellent review articles that have identified the potential critical gaps [98,134,135].

Environmental policy efforts to ensure the judicious use of antimicrobial drugs in agricultural settings are generally focused on minimising or restricting the application of drugs used to treat human infections as growth promoters. According to Codex Alimentarius, growth promotors are defined as the use of antimicrobial drugs to enhance the efficiency of animal feed to increase the rate of weight gain. Therefore, antimicrobial drugs are administered for better feed conversion efficiency and increased growth rates [76]. The UK Joint Committee on the Use of Antibiotics in Animal Husbandry released the Swann Report suggesting a ban on antimicrobial drugs that are shared between animals and humans. Different countries have responded differently to this serious issue and it could take decades to turn around the damage if governments take robust actions [136]. In 1995, Denmark banned the use of two medically important antimicrobial drugs, namely virginiamycin and avoparcin, as growth promoters, whereas Sweden had already banned the use of all antimicrobial drugs as growth promoters in 1986 [137]. The exploitation of medically important antimicrobial drugs (as a patent by the FDA) for growth promotion was banned on 1 January 2017 in the USA. Data from the World Health Organisation for Animal Health (OIE) suggest that in 2015 more than 70% of countries that responded to the assessment did not allow the use of antimicrobial drugs for growth promotion [95]. The OIE and FDA as well as British and US veterinary medical associations have described the concept of rational, prudent or judicious use of antimicrobials in animal agriculture [136].

7. Antimicrobial resistance surveillance

The identification of complexities and measurement of AMR has become a major concern during the study of soil, water and other environmental samples for practical or diagnostic surveillance [138]. Generally, two approaches are used for AMR surveillance, including a culture-based approach and a metagenomic or molecular approach. Currently, nanotechnology-based approaches are also gaining momentum for AMR surveillance. These approaches are briefly mentioned below.

7.1. Culture-based approach

The WHO Global Action Plan for Antimicrobial Resistance has been developed to guide efforts on AMR surveillance and research [139]. In this context, ESBL-producing *E. coli* have recently been suggested as a potential candidate for monitoring AMR owing to (i) quick detection through established methods for high-income countries (HICs) and low- and middle-income countries (LMICs), (ii) quantification of variations in occurrence and prevalence between HICs and LMICs, and (iii) the ability of comparative evaluation of food and medical samples in addition to environmental monitoring [140]. Several other culturable bacterial groups, including Gram-positive and Gram-negative bacteria, have also been recommended for environmental monitoring worldwide. One critical drawback associated with culture-based approaches is the aforementioned fact that non-pathogenic environmental bacteria that are not cultured can represent important resistance reservoirs to complicate the detection process [141].

7.2. Molecular approaches

Nucleic acid-based approaches involving DNA sequencing and metagenomics facilitate the determination of ARGs within and between environmental compartments. However, such methods are expensive and require centralised operational facilities. Nucleic acid-based approaches such as nanopore sequencing are highly useful since they can probe environmental resistomes that are not feasible via culturing methods. Molecular methods have characterised the resistomes of diverse environmental samples to elaborate the potential linkage between faecal contamination and resistance dissemination [142]. Such facilities can comprehensively analyse LMICs and HICs but are not cost effective. However, whether an ARG detected by metagenomics is sited alone in a bacterial genome or as a part of a multidrug resistance cassette (such as on a plasmid) cannot be easily identified from metagenomics data alone [143]. Such issues can be partially resolved by parallel culturing of the same organisms, selection of specific phenotypes, and determining the genetic context of specific genes. In contrast to metagenomics, a better-targeted quantitative PCR (qPCR) method is useful for tracking specific ARGs in the systems where ARGs have already been established. qPCR can also be used for known ARG biomarkers [144].

7.3. Nanotechnology-based approaches

Biosensors based on nanomaterials are increasingly being explored and have a significant advantage due to the low cost of rapid detection of high selectivity ARGs and field applications [145]. For example, gold nanoparticles (AuNPs) have been created for the sensitive detection of *mecA* genes. In this approach, oligonucleotide-functionalised AuNPs were designed to hybridise to the *mecA* gene fragment. The presence of the *mecA* gene in the sample was then based on the change in the suspension absorption spectrum owing to the agglomeration of nanoparticles [146]. Liu et al. developed a biosensor based on oligonucleotide-functionalised AuNPs that was completely immobile on a glass carbon electrode, thus expressing a further benefit over optical methods [147]. One Health initiatives promote collaboration between stakeholders in human and veterinary medicine to conduct full genome sequencing-based monitoring of humans and animals in hospitals and farms [2].

8. Mitigation strategies: global action plan to counter antimicrobial resistance

Numerous European countries have taken initiatives to reduce the incidence and spread of antibiotic resistance by ensuring the judicious use of antimicrobial drugs. Recently, the FDA has released plans to monitor AMR issues [22]. Countries that have developed integrated national plans have effectively overcome AMR. Key factors include the careful use of antibiotics, monitoring of antibiotics through a 'One Health' approach, progress in healthcare systems, development of health insurance policies, limited drug promotion, coherent disease control policy and community stewardship plans

[21]. Moreover, these strategies require a good deal of patience, time to organise and sufficient funds from government authorities. Global agencies such as the WHO Global Antimicrobial Resistance Surveillance System (GLASS), CDC, Food and Agriculture Organization (FAO) and OIE are working hard to control antibiotic resistance. Other programmes addressing the global threat of antibiotic resistance include the Global Health Security Agenda (GHSA) and the Antimicrobial Resistance Action Package (GHSA Action Package Prevent-1) [148,149].

Diagnostics is also another urgent scenario especially in developing countries as they still rely on traditional microbiological tools to identify bacteria. Such gaps could be bridged by developing personalised medicines based on modern and improved molecular diagnostic tools for needed antibiotic therapy. A 'One Health' approach could be a very effective tool to study the human–animal interface and design novel screening tools. This subject is of unique importance and must be taken seriously because the mechanisms of AMR transmission have been identified between all components (humans, animals and environments) of single health [150]. Imprudent and irrational use of antibiotics is a key factor in AMR, particularly in LMICs. Antibiotics are imprudently used for several reasons such as patient's satisfaction with the doctor's prescription, inadequate information about antibiotics, improper diagnosis especially in developing countries, and disturbing adversities for physicians by the pharmaceutical industry. A shortage of new antibiotics can complicate the investigation of this part of the problem [151], therefore advancements in the development of antibiotics, combination therapy and technical innovation are indispensable [152]. Major aspects of future research are to identify the influence of organisational anthropogenic inputs, the role of various factors in the development of AMR, the impact of resistant bacteria on human and animal health, and major technological, social and economic initiatives for the mitigation of environmental antibiotic resistance.

9. Conclusions and future perspectives

The environment can act as a reservoir of antibiotic-resistant bacteria and ARGs or as an arena for the evolution of resistance. Wild animals and livestock can potentially serve as reservoirs and melting pots for bacterial drug resistance. Therefore, it is the most common route of transmission of resistant bacteria associated with faecal matter in water, sewage infrastructure or organic fertilisers. Foodstuffs of animal origin may also be considered as a potential source of exposure to multidrug-resistant enteric pathogens. There must be a well-established model system for risk assessment of infections and deaths resulting from exposure to antibiotic resistance arising from environmental antibiotic residues. Such a type of model system would be very powerful in mitigating and monitoring environmental sources of resistance emergence and its transmission. This would help in reducing the outcomes of environmental antibiotic residue exposure to humans and decreasing the associated health risks. There is also a huge challenge in the cultivation of environmental bacteria that restricts the determination of resistance factors carried by the host. These limitations can be overcome by employing PCR-based molecular techniques. In addition, there are technological limitations that constrain our understanding of environmental resistance and their horizontal transmission among bacteria. All relevant decision-makers must be aware of the AMR threat and should apply the regulatory frameworks in agriculture, pharmaceutical manufacturing and aquaculture to minimise the development of antibiotic resistance. Developed and developing countries facing various challenges must be cognisant to develop the policies and research network for the mitigation of AMR. Development of methods with increased sensitivity towards rare resistance determinants without cultivation as well as improved ability to culture a broad range of species could expand our opportunities to achieve a deeper understanding of the associated risks.

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Competing interests

None declared.

Ethical approval

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